Reviewer Recommendation and Comments for Manuscript Number JBI-21-293

Predicting Drug Blood-Brain Barrier Penetration with Adverse Event Report Embeddings

Original Submission
Zhen Gao Reviewer 4

Back | Edit Review | Print | Submit Review to Editorial Office

Recommendation: Major revisions necessary

Custom Review Question(s):	Response
1. TECHNICAL CORRECTNESS	Acceptable
2. NOVELTY/ORIGINALITY	Good
3. REFERENCE TO PRIOR WORK	Good
4. QUALITY OF ART	Acceptable
5. QUALITY OF EXPERIMENTAL RESULTS	Acceptable
6. APPROPRIATENESS TO JOURNAL	Acceptable
7. IMPORTANCE TO THE FIELD	Acceptable
8. ORGANIZATION AND CLARITY	Acceptable
9. LENGTH	Acceptable

Reviewer Comments to Author

General comments:
The authors of this manuscript adopted distributed neural embedding learned from drug AER-based similarity plus 2 chemical features to predict permeability of the blood-brain barrier (BBB) to drugs. To my knowledge, This approach has some novelty, but there are quite a few concerns left.

Major:
The material/workload is Insufficient. Only Figure 2 and Table 1, 2, 3 are the results presented. Much more work is needed to make this manuscript a robust one:
1) The authors adopted a 6 BBB+ drugs panel for the base of similarity comparison. But why these 6 drugs? Are these drugs gold-standard for characteristics of BBB+ drugs or these 6 drugs have the best performance? If it's the former case, the authors should provide evidence; if it's the latter case, the author should provide the process of how these drugs are selected.

2) A 500-dimensional distributed representation is used as a feature in the research, but I have no clue what's the data look like, the authors should provide such data and explain them.

3) PCA is NOT a feature selection tool. It's just a tool to distribute variability on the orthonormal basis. Scientists could keep any level of variability by including different PCA component numbers. The authors should use real feature selection tools such as lasso, elastic net, or random forest.

4) The author essentially presented only one machine learning method. They could try quite a few more for performance comparison (for example, neuron network).

5) Dose the Aer2vec+ have the possibility for P(permeability|ADE)?

6) The author showed performance on two datasets with known permeability. But authors should provide predictions for some drugs/compounds without earlier knowledge of permeability, then show these compounds have evidence/support from recent experimental/observation/animal work. The aim of scientific prediction is always to predict unknown compounds, e.g. for drug screening. The performance on known datasets is just an indication, not scientific aims.

7) Finally, where is the code? Code is critical for reproduce their results, performance evaluation, and information sharing.

Reviewer Confidential Comments to Editor:

Dear Editor:
The novelty of this draft is ok, but I prefer them add more works.
The current version has very small workload.
There are many work need to be added.

Best,
Zhen

Transfer Authorization	Response
If this submission is transferred to another publication, do we have your consent to include your identifying information?	Yes
If this submission is transferred to another publication, do we have your consent to include your review?	Yes

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